Handbook of Research on Interdisciplinary Perspectives on the Threats and Impacts of Pandemics

Şahver Omeraki Çekirdekci
Haliç University, Turkey

Özlem İngün Karkış
Doğuş University, Turkey

Suna Gönültaş
Doğuş University, Turkey

A volume in the Advances in Human Services and Public Health (AHSPH) Book Series



Published in the United States of America by
IGI Global
Information Science Reference (an imprint of IGI Global)
701 E. Chocolate Avenue
Hershey PA, USA 17033
Tel: 717-533-8845
Fax: 717-533-8661
E-mail: cust@igi-global.com
Web site: http://www.igi-global.com

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Library of Congress Cataloging-in-Publication Data

Names: Omeraki Çekirdekci, Sahver, 1979- editor. | Karkis, Ozlem Ingun, 1979- editor. | Gonultas, Suna, 1980- editor.
Title: Handbook of research on interdisciplinary perspectives on the threats and impacts of pandemics / Sahver Omeraki Çekirdekci, Ozlem Ingun Karkis, and Suna Gonultas, editor.
Description: Hershey PA : Business Science Reference, [2022] | Includes bibliographical references and index. | Summary: "Beginning as a health threat, the pandemic has led the way to economic, social, psychological, political, and informational crises necessitating the examination of the phenomenon from different academic disciplines with this book giving valuable insights on the threats, disruptions, and responses to the new normal created due to the COVID-19 pandemic"-- Provided by publisher.
Identifiers: LCCN 2021024010 (print) | LCCN 2021024011 (ebook) | ISBN 9781799886747 (hardcover) | ISBN 9781799886761 (ebook)
Subjects: LCSH: COVID-19 Pandemic, 2020---Social aspects. | Epidemics--Social aspects.
Classification: LCC RA644.C67 I5788 2022 (print) | LCC RA644.C67 (ebook) | DDC 362.1962/414--dc23
LC record available at https://lccn.loc.gov/2021024010
LC ebook record available at https://lccn.loc.gov/2021024011

This book is published in the IGI Global book series Advances in Human Services and Public Health (AHSPH) (ISSN: 2475-6571; eISSN: 2475-658X)

British Cataloguing in Publication Data
A Cataloguing in Publication record for this book is available from the British Library.

All work contributed to this book is new, previously-unpublished material. The views expressed in this book are those of the authors, but not necessarily of the publisher.

For electronic access to this publication, please contact: eresources@igi-global.com.

Chapter 14
Towards a Restructured Theorization of Development and Adaptability for the Post-COVID-19 Era:
The Stra.Tech.Man Approach

Charis Vlados
https://orcid.org/0000-0003-2509-6961
Democritus University of Thrace, Greece & University of Nicosia, Cyprus

Dimos Chatzinikolaou
Democritus University of Thrace, Greece

ABSTRACT

It becomes progressively understood that the health crisis (and the deriving socioeconomic crisis) of COVID-19 drives the global system towards radical transformations. The successful adaptation of the various socioeconomic actors in this emerging reality probably constitutes the most significant challenge ahead. This chapter suggests a restructured theorization of socioeconomic development for the post-COVID-19 era, focused on the prerequisite organizational adaptation. This conceptual research examines fundamental evidence concerning current developments globally, arriving at a repositioning in evolutionary and interdisciplinary terms by adopting the strategy-technology-management synthesis to change management and innovation (Stra.Tech.Man).

INTRODUCTION

COVID-19, according to several analysts, drives the acceleration of the ongoing fourth industrial revolution; hence, past central occupations in traditional industries are rapidly decaying, and others are emerging (Bonilla-Molina, 2020; Bragazzi, 2020; Czifra & Molnár, 2020). In the interregnum, based on

DOI: 10.4018/978-1-7998-8674-7.ch014

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the experience from the earlier three industrial revolutions, social unrest, rise in absolute poverty, and economic inequality seem inevitable consequences (Schwab & Malleret, 2020). Thereby, in this crisis of globalization, symptoms such as neo-Luddism (McKay, 2020), political extremism (Cox, 2017), and intolerance might appear as the new normal, at least until the advent of a sufficiently stabilizing new phase of global development (Vlados, 2019b).

COVID-19 arrived at a moment when the global system was already within a fundamental readjustment and restructuring. The earlier global capitalism crisis, which manifested across all socioeconomic systems in a ripple effect fashion, was outbroken due to unsustainable financial derivatives that "polluted" the entire world economy (Andrikopoulos & Nastopoulos, 2015; Gup, 2010). However, this external cause of the crisis internally carries—and from an Engelian perspective (Engels, 1873)—a quantitative accumulation in generic socioeconomic terms that inevitably led the global system to a profound qualitative change. This crucial qualitative transformation was the springboard for socioeconomic, institutional, and political alterations altogether. One of the most significant restructuring forces was (and still is) BRICS (Brazil-Russia-India-China-South Africa), which increasingly challenges the Western-inspired institutions (Vlados, 2020b).

Specific past analyses that shed light on the global system's evolutionary development help examine global capitalism's future. According to the neo-Schumpeterian research of C. Perez (1983), the world economy moves forward with specific techno-economic paradigm shifts that require five to six decades each time, spurred by waves of infrastructure investment that induce the attainment of full growth potential because the paradigm becomes generalized and diffused in institutional and structural terms. It can be said that Perez's theory interprets today's call for digital transformation, which is an ongoing goal that governments and firms alike strive to accomplish in the post-COVID-19 era (European Council, 2020; Soto-Acosta, 2020; Winarsih et al., 2021). In K. Schwab's point of view (Schwab, 2016), the world has entered the fourth industrial revolution that refers to the fusion of technologies that blurs the physical, digital, and biological spheres. This blurring that is called diffusion of "cyber-physical" systems in the fourth industrial revolution terminology appears to be accelerating in the post-COVID-19 era based on the spread of digital means to conduct business, education, and other daily tasks in a vast and growing array of industries (Mhlanga & Moloi, 2020; Savić, 2020).

Thus, it appears that the global socioeconomic system's mutation is accelerating after this pandemic crisis in various emerging directions, and this happens for two reasons. First, COVID-19 effects caused an exogenous shock that has resulted in radical macro-level shifts. Second, these COVID-19 repercussions became endogenous, generating multiplying impacts that continue to influence the global equilibrium in social, economic, and political terms. By considering these accelerating developments, the question is how socioeconomic organizations can deal with this evolutionary turmoil and take advantage of the new phase of global transformation that unfolds. As it seems, based on available data, various industries offer significant development opportunities nowadays. Indicatively, e-commerce retail sales in April 2020 (one of the pandemic's picks worldwide) recorded an increase of 30% in the European Union; this statistic is significant, especially if we consider that total retail sales in the same month fell by 17.9% (OECD, 2020). The recalibrations in the education sector also are radical since, according to UNESCO et al. (2020), most national governments took at least one measure or policy during 2020 to facilitate access to distance learning for social groups that are facing more intensely the danger of potential exclusion from formal education structures. This report also notes that three out of ten lower-income countries did not take such measures (which illustrates the competitiveness gap in overall socioeconomic terms; Peneder, 2017; Vlados & Chatzinikolaou, 2020). However, this adverse development does not mean

that there are no chances for future growth. A World Economic Forum (2020) report presents a survey on various business leaders on whether their firms have introduced new digitalization policies; 80% of these reported that they have already taken such actions, expanding their firms' remote working and automation processes. Therefore, we observe diverging prospects in the development and adaptability of different socioeconomic organizations and spatial systems.

This research is structured according to the basic principles of conceptual studies since it critically approaches theories and concepts of the current mutational crisis, aiming to develop a reframed theorization of organizational adaptation (Jaakkola, 2020). More specifically, novel insights will result by combining facts concerning the global socioeconomic crisis of COVID-19 with approaches that contrast monolithic economic growth with the more comprehensive and holistic economic development concept.

In terms of the interpretive approach followed, fundamental elements of the new global framework after COVID-19 are examined in the next section, focusing on significant macro-level phenomena. In the subsequent section, a repositioning of socioeconomic development is attempted, ending with the need for an evolutionary, historical, and interdisciplinary perspective. In the concluding section, final remarks are made by suggesting a new framework for understanding organizational adaptation in the post-COVID-19 era, based on contributions to organizational evolution and the "Stra.Tech.Man" approach to innovation and change management (strategy-technology-management synthesis).

FOCUSING ON THE POST-COVID-19 ERA NEW GLOBAL FRAMEWORK

A stroke is defined as the condition where the bloodstream suddenly stops in a human brain's area (Collin, 1998). In the absence of blood flowing to and from the brain, the affected brain cells slowly die and cannot be regenerated, causing disabilities to appear and related symptoms. When the initial phase of the stroke is over, the recovery effort begins. The rehabilitation measures focus on neuro-muscle training and related practical tasks to reinforce the patient's self-service dexterities. The recovery program's effectiveness depends on the patient's current health conditions, including chronic diseases or morbidity factors. Even though we are not doctors in medicine, we find the "stroke metaphor" the most appropriate to describe today's global developments. It can be said that the world economy seems to be suffering nowadays from a stroke that has manifested in all national socioeconomic systems, leading to a spectacular rise in unemployment, especially in the hardest-hit industries. Moreover, specific ramifications in various socioeconomic niches caused by the COVID-19 pandemic can reinforce the argument that this unprecedented crisis is structural and paradigmatic:

- The United Nations (2020), in a report addressing the critical response to the COVID-19 crisis on the humanitarian level, has seen the rise of extreme poverty, unemployment, and new famines as a definitive possibility. UNCTAD (2020), a body responsible for trade and development issues within the United Nations, has declared an extreme disruption in global value chains, reflected in an extraordinary fall of global foreign direct investment of approximately 40%. At the structural level of regions, the Food and Agriculture Organization (FAO, 2020) of the United Nations has stated that rural areas cannot cope as effectively with the crisis since 80% of the total 734 million people can be found in such places, and particularly in agriculture, living in conditions of extreme poverty before COVID-19.

- In a similar report addressing the devastating effects of COVID-19, the World Trade Organization Director-General, R. Azevêdo (2020), has suggested an inevitable plunge of world merchandise trade volume amid and after the pandemic.
- As per the International Labor Organization's calculations on the imprint of COVID-19 on work (ILO, 2020), the results are equally worrying since four hundred million full-time jobs were lost worldwide during the second quarter of 2020. The ILO (2021b) also has examined the implications of remote working, finding that only 18% of workers live in countries with infrastructure that can support effective teleworking, while only one in four jobs can be carried out solely with digital means and remotely in developed economies and only one in six internationally. As per the topic of wages in the post-COVID-19 era, the ILO (2021a) has distinguished the most influenced industries, primarily wholesale and retail trade, accommodation and food services, and other economic sectors where women are mainly employed, and the latter has potentially disastrous effects in terms of social inequality and upset.
- In a series of related reports and press releases, the World Bank's conclusions have signaled a warning, based on the fact that the sharpest decline in remittances globally in recent history has been now recorded (World Bank, 2020a). It has also forecasted specific scenarios for the possible rebound of the world economy in 2021, ranging from downside to baseline and upside (World Bank, 2020b). The World Bank concluded that the COVID-19 pandemic arrived when the global economy was already vulnerable—and despite the unprecedented measures taken to support the various communities and national economies, this recession will surpass all other recessions after World War II.
- On behalf of the International Monetary Fund (IMF, 2020), the World Economic Outlook has readjusted at least twice its predictions for the declining global output in 2020. This move demonstrated the fear for repeated social distancing measures due to a novel outbreak, implying that this crisis is "like no other" in human history in magnitude and complexity.
- On the future of jobs, a recent report by the World Economic Forum (2020) has illuminated the working environment's expected profile by 2025. According to the report's authors, COVID-19 has worsened existing inequalities, which will continue to increase due to the spread of modern technology and digital means for working. These digital inequalities will affect the most vulnerable population, such as the long-term unemployed, the unskilled, women, the new entrants in labor markets, and those employed in less competitive and well-paid jobs.
- Finally, implications at the industry level need to be mentioned because they can depict structural imbalances worsened by the COVID-19 crisis. According to the International Civil Aviation Organization (2020), international airline passengers have declined at figures as low as in the early 1990s. The International Energy Agency (2020) has estimated in its world energy outlook report that energy demand fell in 2020 by 5% and energy investment by 18%. The pandemic's adverse effect on industries such as infrastructure, catering and media, broadcasting, the sports industry, and many others is equally significant, disrupting social development in the foreseeable future (United Nations, 2020b).

What is the structural and long-term meaning of these developments? Matthewman and Huppatz (2020) have proclaimed this pandemic crisis as a gateway between one world and the next. Other scholars have suggested that these trends spur the end of the last phase of globalization as we know it thus far (Altman, 2020; Gorynia, 2020). Overall, we see dramatic changes influencing all individuals and

socioeconomic systems worldwide, with negative macroeconomic patterns expected to bring significant changes in our social symbiosis in the years to come. An exacerbation of social exclusion, especially for the weakest parts of the population in developed and developing countries, can lead various localities to vicious circles of institutional under-development (Acemoglu & Robinson, 2012). The problems posed by the current COVID-19 crisis and the ones that will appear after the immediate health crisis (the "post-COVID-19 era") affect all socioeconomic systems and actors since no industry and broader socioeconomic configuration can remain unaffected by the dynamics of globalization (Boyer, 2019; Puig & Urzelai, 2019). In this context, we think that the arising critical question is "what does socioeconomic development mean in the post-COVID-19 era." The following section examines theoretical restructurings for interpreting the phenomena appearing in the ongoing post-COVID-19 transition.

CONTROVERSIES AND MULTIPLE PROBLEMS FOR POST-COVID-19 SOCIOECONOMIC DEVELOPMENT

Research into the underlying causes of economic growth and development can be traced back to the works of J. Schumpeter (1942) and N. Georgescu-Roegen (1971), though seeds of this distinction can also be found in the works of A. Smith (1776), J.S. Mill (1848), K. Marx (1867), and other classical economists. Evolution and economic development in these works refer to more profound phenomena than the mere accumulation of quantities. In Krugman's (1996) perspective, "economic development" and "economic growth" emerged as new subdisciplines in the post-WWI era, with growth economics stemming from achieving full employment in market economies. On the contrary, development economics is derived from the analysis of economic growth in weaker and less-developed societies. Ruttan (1998) argued that growth economists believe that development economics is imprecise and ambiguous, while development economists assume that their "counter-partners" occupy themselves only with market information to determine interest rates and other types of economic quantities. From an evolutionary perspective, for F. Perroux (1969), economic development is inextricably related to evolution and refers to irreversible shifts in events and processes linked to one another rather than being a series of random elements. Overall, the development process involves a much broader framework of structural changes—beyond simple growth—referring to the entire upgrade in the production and consumption model of the various societies, expressed in higher levels of efficiency, well-being, justice, and democracy at multiple organizational levels (Conceição & United Nations Development Programme, 2019; OECD, 2018). In this context, by examining literature that concerns the controversy "economic development versus economic growth," we identify at least ten sources that consider economic development a superior methodological approach. We present them below in their chronological succession:

i. Economic development is a more complex issue, and economists have characterized the phenomenon as growth accompanied by changes in the country's structure and sociopolitical environment (Zuvekas, 1980).
ii. Contrary to economic growth that deals mainly with the Gross Domestic Product (GDP), economic development deals with the essential underlying qualities (Brown et al., 1992).
iii. The quantitatively static economic growth approach must not be synonymized with economic development's profound structures (Brinkman, 1995).

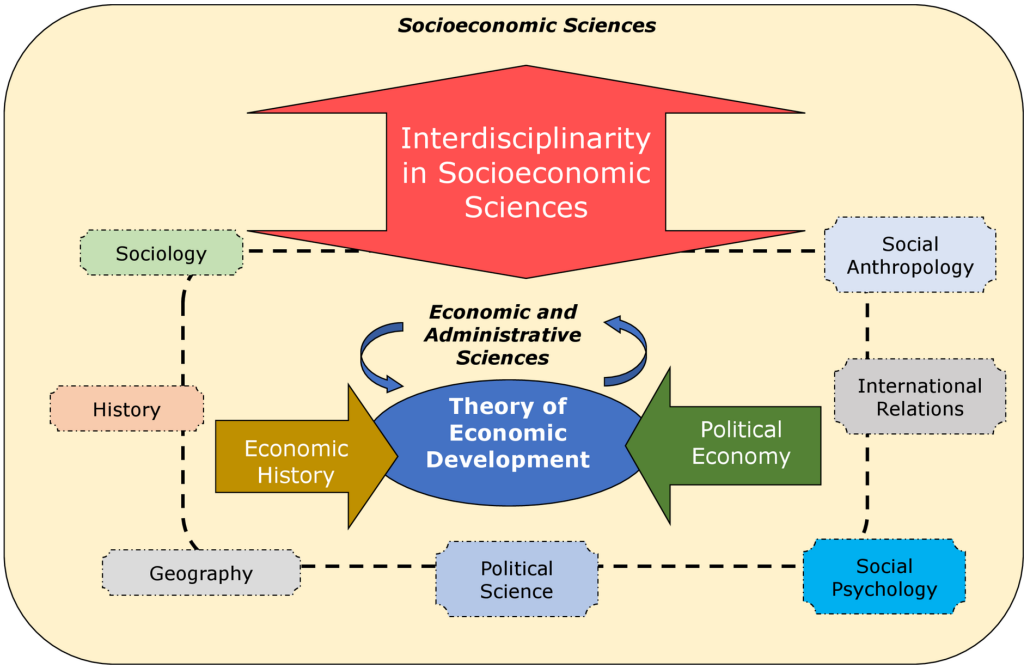
iv. Seeking infinite growth and expansion within a bounded system is a practice of growth economists that can only be unsustainable (Chiras, 1995).
v. Mainstream theories of the past century usually do not distinguish between "growth" and "development," mistakenly treating the two concepts as the same. Development refers to qualitative factors necessary for long-term success (Papanek, 2002).
vi. In the first days of development economics, per capita GDP was the primary tool for considering economic development, leading to confusing development with the less complex economic growth concept (Hosseini, 2003).
vii. Growth assumes that certain factors stay constant from a comparatively static standpoint due to the ceteris paribus hypothesis[1]. A dynamic development approach deals with multiple and successive structural transformations on the economy's foundation (Wang et al., 2008).
viii. Economic development simultaneously considers all facets of socioeconomic co-existence, implying that "development means transforming for the better" and usually from a bottom-up perspective (Peet & Hartwick, 2015).
ix. Despite most scholars agreeing that development refers to something broader than growth, nations continue to appeal to the GDP growth rate as their primary concern (Xu & Liu, 2017).
x. Development and growth cannot be interchangeable because economic growth is expressed in increasing national or per capita income and GDP, whereas economic development relates to improving living standards, reducing poverty, and supporting structural changes to the economy's composition (Nnadozie & Jerome, 2019).

According to these perspectives that suggest the conceptual prevalence of economic development against economic growth and agreeing for the most part with D. Acemoglu (2012) that "economic development is still a major problem for humanity at large and economics as a science," we put forward the need for an interdisciplinary and evolutionary socioeconomic understanding of development in the post-COVID-19 era. Arguing that development economics by nature has a broader outlook (in agreement with the older but still valid contribution by Gillis et al., 1996), three parallel analytical platforms of enrichment seem necessary (Vlados, 2020a):

I. An evolutionary perspective is needed to the economics of development: Economic problems have no "one best way"—linearly and statically—according to traditional analyses that mostly derive from the conventional neoclassical thinking (Nelson & Winter, 1982). Development problems in the post-COVID-19 era are complex and require examining the socioeconomic systems as organic entities that co-evolve within a dense web of interrelationships. This is why research on entrepreneurial ecosystems seems to rise amid the COVID-19 pandemic, striving to propose effective policy solutions that could foster innovation generated by the combined forces of academia-industry-government (Márquez-Ramos, 2021; Mason & Hruskova, 2021; Sahasranamam, 2021).
II. History continually formulates new development dynamics: No "end to history" will ever occur, as implied by earlier theorists (Fukuyama, 1992). Especially in the post-COVID-19 era, already less-adaptive and less-resilient spatial socioeconomic systems, according to their specific idiosyncrasy, will face barriers to inserting themselves into the dynamics of the fourth industrial revolution (Arbolino & Caro, 2021; Iacobucci & Perugini, 2021).

III. Economic and administrative sciences need to move from partial specializations towards an interdisciplinary approach to socioeconomic development dynamics: Scientific theorizing must not deal with the development phenomena from an "autonomous" perspective (Figure 1). Economic development theory in the post-COVID-19 era should function as an interdisciplinary research hub for all socioeconomic disciplines (Curran & Eckhardt, 2020; Wen et al., 2020).

Figure 1. Interdisciplinarity in socioeconomic sciences.
Source: Adjusted from Vlados (2020a)



As it seems, the global community needs a new way of thinking to surpass the current pandemic and face the least possible losses. This new perception of economic and administrative sciences must be refocused on the broader economic development theory and enriched with contemporary political economy and economic history elements. Today's economic science needs a thorough and interdisciplinary way of thinking synthesizing the interests, methods, and findings of different disciplines, such as social anthropology, international relations, social psychology, political science, geography, history, and sociology.

CONCLUSION

This section functions as a counterproposal to organizational adaptation problems for the post-COVID-19 era. Specific fundamental change management models are examined first, followed next by the "Stra. Tech.Man perspective" (strategy-technology-management synthesis).

Thus far, we have examined the drastic multilevel changes brought about by the current pandemic crisis of COVID-19 and the necessary renovation in development economics' theoretical arsenal. Based on the suggested evolutionary and interdisciplinary perspective, we conceive every socioeconomic organization as a living entity that has specific development capabilities that are "imposed" by its external environment and own homeostasis (Geus, 1997; Kelly, 1994; Meyer & Davis, 2003; Vlados, 2019a; Zeleny, 1980). The reference to socioeconomic organizations in this evolutionary and biological-type approach is broad, considering individuals, enterprises, and every other institution in the development process. Evolution for socioeconomic organizations is not an abstract concept. According to Hodgson (2002), we cannot apply all biological features at the socioeconomic level, although general patterns of "Darwinian" origin appear as long as variation, selection, and inheritance occur. Foster (1997) also favors a "Lamarckian analogy" for socioeconomic organizations instead of the Darwinian, arguing that behavioral characteristics gained from experience formulate the specific environments and, thus, the biological natural selection is not the primary adaptation context.

Nevertheless, the question is still open: What can the socioeconomic organizations do for adapting and surviving in emerging environments, especially if these are turbulent such as the post-COVID-19 era? According to the "change management" theory developed over the past years, socioeconomic organizations can help themselves if they build effective mechanisms for managing the change that will come up to the surface sooner or later. It seems that even today, we can draw valuable elements from older and more recent approaches, which help us gain a better understanding of the challenges concerning change management at all levels of activity in the emerging post-COVID-19 conditions. Some of them are the "unfreeze-change-refreeze" technique (Lewin, 1943), the "PDCA cycle" (plan-do-check-act) by Deming (1986), the approach of change and leadership by Kotter (1996), as well as the management of chaos by Kotler and Caslione (2009). More specifically:

- Lewin's (1943) change management technique of "unfreeze-change-refreeze" refers to a three-staged change. In the unfreeze stage, a motivating event creates a need for change. Once the organization decides to change, then a change management strategy must prepare employees for the transition. In the change stage, the actual changes to the organizational structure, business practices, staffing, or other areas begin and diffuse throughout the company. In the refreeze stage, the second-stage changes become normalized in the organization's day-to-day activities. This process can be slow, depending on the scale of the implemented changes, because it can take a long time for the company members to learn and assimilate new practices.
- Deming (1986) popularized the "PDCA cycle" approach, showing that there is a cycle for implementing change that, when followed and repeated, would lead to repeated improvements in the applied process. The first part refers to the "Plan" stage, where a well-defined project plan supplies the general operational framework, ideally reflecting the organization's mission and values and mapping efficiently the project's goals and the best ways to meet them. The "Do" phase involves setting the plan in motion, getting all personnel involved in the project through training, recording insights and data for future evaluation. The "Check" stage ensures that the organization still looks to fulfill the project's goals, supplying a comprehensive review of the project to allow the future address of successes and failures and suggest future adjustments. Finally, the "Action" phase refers to corrective actions taken in the last step. Overall, this change management process can find past mistakes and repeat a cycle that might lead to better results.

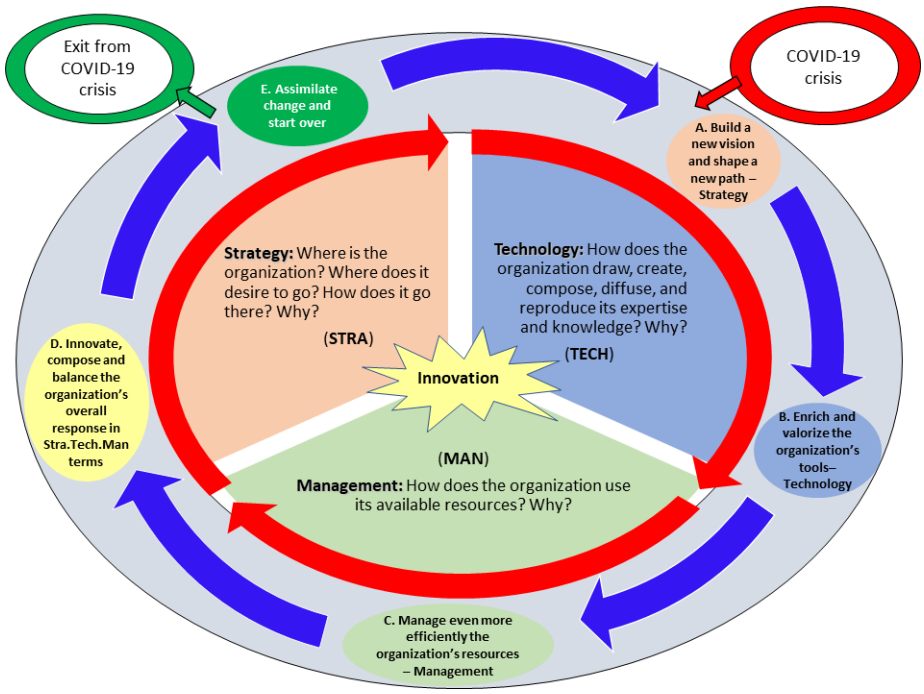
- Kotter's (1996) approach to change focuses on the organization's leaders, who must efficiently understand and implement the plan. This eight-step model begins with creating a sense of urgency, meaning taking everyone out of their comfort zones. Then, it goes on with forming a guiding coalition by creating a good project team. Step three refers to developing an inspired vision to ensure that everyone understands the plan's aims. Step four conveys the new perspective by communicating it to the broader organization's shareholders and stakeholders. Step five refers to empowering others to enact the vision; in the sense that the leader needs to make sure that all employees involved have the required skills and resources to keep pace. Step six is about generating short-term wins by breaking down the project into smaller short-term goals to boost morale. Step seven refers to sustaining the vision's acceleration, meaning that change is always a slow process that needs cultivation within the organization's underlying values, culture, and goals. The last step is instituting permanent change, suggesting that leaders need to anchor and embed change within the organization's core and culture. Kotter concludes that leaders need to monitor this whole initiative to consolidate change.
- Kotler and Caslione (2009) present what chaos means in global business reality, suggesting practical solutions to (mostly larger) organizations to fight unprecedented situations. They note that this new era requires an entirely new organizational conception and action. Today's way of existing turbulence requires new practices so that businesses prosper in the age of chaos. Companies must make significant efforts to avoid duplication of capabilities, have simple contracts based on long-term trust, and have supplier and distributor rating systems that are easy to understand and give immediate feedback. They also must have suppliers and distributors proactively suggest modifications that would improve products and select suppliers based on substantial input from cross-functional teams within the company. Kotler and Caslione add that the co-location of facilities promotes superior communication between significant suppliers and distributors, while firms need to improve supply management and embrace more single-sourcing or reduced sourcing relationships. They also argue that waiting too long to drop relationships with suppliers and distributors who are poor or marginal performers, together with training the organization's suppliers and distributors to reduce operating costs and increase sales, might be best practices. Finally, the authors suggest companies investing in various methods to improve communications with suppliers and distributors.

Upon these foundations and borrowing elements from evolutionary theorizing in economics, the approach of evolutionary management that has been developing over the recent years is remarkable towards the direction of dealing with today's complex socioeconomic reality because it examines as central concepts the dimensions of organizational adaptability and flexibility (Endres, 2018). This new paradigm of evolutionary management allows conceiving more comprehensively than the earlier approaches the present-day compound nature of administrative systems, which is the product of continuous evolutionary processes and innovations that result in progressive performance improvements (Donaires & Martinelli, 2019). Various authors focus on different organizational aspects in the related literature that studies how the management perspective transforms in the post-COVID-19 era substantially. For instance, Liguori and Pittz (2020) observe the need for aggressive and cost-efficient moves to adequately manage today's disruption. Also, Verma and Gustafsson (2020) suggest the need for modifying supply chain structures for companies to survive in the aftermath of COVID-19.

To sum up, these approaches create a sufficiently developed theoretical impetus for realizing today's challenges. Nevertheless, in the context of this theoretical substratum, various insufficient conceptions and practices tend to reproduce themselves, characterized by a relative absence of the required synthesis and integrated solutions to problems (Scharmer & von Ameln, 2019). Change management efforts often focus on technology issues one-dimensionally, under-evaluating the strategic and managerial implications to effective organizational adaptation. Furthermore, approaches exhausted in everyday management problems tend to omit today's prerequisite strategic repositioning and continuous technological advancement for integrated organizational progress. Thus, a reconceptualization and enrichment of today's change management seem critical to exceeding these analytical weaknesses; toward a more synthesizing and holistic direction, which emphasizes the interdisciplinary approach for organizational adaptation in the emerging conditions—social, technical, psychological, and economic dimensions, perceived as an organic whole within the various organizations.

More specifically, the "Stra.Tech.Man" approach (strategy-technology-management synthesis), a method for diagnosing the management philosophy that we have developed in the recent past, we think could offer more adequate answers in terms of adaptability and innovativeness for all socioeconomic organizations in the post-COVID-19 era (Vlados, 2006, 2016; Vlados & Chatzinikolaou, 2019a). The strategy-technology-management approach to innovation and change management deconstructs the socioeconomic organization as a "physiologically evolving" entity, following how it synthesizes the inner dimensions of strategy, technology, and management. These three Stra.Tech.Man spheres are analyzed according to different sets of compound questions, which lie at the root of every innovation effort (Figure 2):

Figure 2. The Stra.Tech.Man perspective of change management for the post-COVID-19 era.
Source: Adjusted from Vlados (2016)



i. Strategy: "*Where is the organization? Where does it desire to go? How does it go there? Why?*"
ii. Technology: "*How does the organization draw, create, compose, diffuse, and reproduce its expertise and knowledge? Why?*"
iii. Management: "*How does the organization use its available resources? Why?*"

This "Stra.Tech.Man approach" is substantially different from the traditional models for change management since it primarily emphasizes synthesizing the three discrete spheres that co-evolve within every socioeconomic "organism" (strategy, technology, and management). Furthermore, it does not succumb to the usual analytical myopia of other approaches that perceive change as the one-dimensional outcome of one sphere (By, 2005; Sharifi & Claxton, 2015). In practice, the Stra.Tech.Man approach can be used in all organizations, whether big or small, private or public (or mixed), for-profit or non-profit, locally or cross-spatially focused. In this sense, it seems capable of offering a fertile platform for dealing with today's challenges since the post-COVID-19 profoundly-altered external organizational environment can be only perceived with a sufficiently integrated, cohesive, holistic, and interdisciplinary comprehension. According to the Stra.Tech.Man approach, only the synthesis of the three spheres (strategy-technology-management) can lead to organizational innovation, which is the precondition for s viable way out of the pandemic for all actors, irrespectively of their location and functional specialization.

From the Stra.Tech.Man perspective, the mutation of organizational physiology is at the center of any development and innovation process in all socioeconomic systems (individual, local, regional, national, supranational). Furthermore, from a biological viewpoint, the external environment is continuously transformed and restructured by the co-evolving internal changes in the different symbiotic organizations. These internal changes correspond to the generation of innovation, replacing the old and ineffective status quo in any organizational system by leading to the change management imperative. In the Stra.Tech.Man approach, we can distinguish five phases of change management, which can function as general principles for all socioeconomic organizations:

A. Build a new vision and shape a new path—The strategy sphere (Hill & Jones, 2004; Yaeger & Sorensen, 2009).
B. Enrich and valorize the organization's tools—The technology sphere (Abbott, 2013; Abdelnour-Nocera, 2013).
C. Manage the organization's resources even more efficiently—The management sphere (Carmichael, 2011; Senge, 1990; Tobin & Pettingell, 2008).
D. Innovate, compose, and balance the organization's overall response in synthesizing Stra.Tech.Man terms—The strategy-technology-management innovative integration (Fonseca, 2002; Szirmai et al., 2011).
E. Assimilate change in a holistic way and start over (Hayes, 2018; Raina, 2018).

Therefore, the pandemic crisis inevitably causes the organizational system to enter the dynamics of the Stra.Tech.Man change management cycle. In the next step, the generated innovation constitutes the motor for exiting this and any other crisis. This approach can be a powerful tool for the necessary adaptiveness and innovativeness of socioeconomic organizations in the post-COVID-19 era. It offers a lean and operational framework, which allows diagnosing the "patient" over time, leaving the field open for new syntheses at multiple organizational levels. This compound viewpoint also approaches socio-

economic development from an interdisciplinary perspective because it examines the actors through a cross-thematic prism (in symbiotic and evolutionary terms).

In conclusion, we highlighted the post-COVID-19 era as the recent phase in the evolution of global capitalism, which accelerates multilayered alterations at the institutional level. This acceleration of the fourth industrial revolution might leave at the lower added-value levels various localities and industries (Rodríguez-Cohard et al., 2020; Sadeghloo et al., 2018). Also, this article suggested that to sufficiently understand the evolution of global capitalism and the various organizational adaptations, a new theorization of development must exclude any conventional linearity and serial thought that usually appears as a "standard paradigm" in economic and administrative sciences, focusing monothematically on growth issues. On the contrary, we argue that socioeconomic development in broad terms (especially nowadays) requires interdisciplinary, historical, and evolutionary responses (Mol & Hardon, 2020; Reiss, 2020). In this context, all socioeconomic organizations (firms, state mechanisms, and social institutions) will need to readjust their existing managerial, technological, and strategic priorities to effectively adapt to the emerging conditions (Ansell et al., 2020; Vlados & Chatzinikolaou, 2019b).

This analysis finally has implications for future research. This need for rethinking organizational adaptation in the post-COVID-19 era led to the reformulation of the Stra.Tech.Man approach to change management and innovation (strategy-technology-management synthesis). This evolutionary perspective can function as a fundamental tool for diagnosing the physiology of diverse socioeconomic organizations, whose competitiveness reinforcement can lead to virtuous development spirals for the socioeconomic systems that attract them. Therefore, the Stra.Tech.Man synthesis could also be the basis for policies to support local development—and the progress of this research over the past years leads us to this conclusion (Vlados et al., 2018, 2019). At the same time, the Stra.Tech.Man approach seems capable of taking various forms, most notably as a "scorecard" that can diagnose innovativeness in specific firms and industries (Vlados, 2021). Overall, the Stra.Tech.Man analysis appears to have the potential to be implemented in a wide area of cases in the future (in diverse industries and spaces).

ACKNOWLEDGMENT

We want to express our sincere gratitude to Dr. Andreas Andrikopoulos, who offered valuable comments during this paper's writing.

REFERENCES

Abbott, L. F. (2013). *Technological development in industry: A business-economic survey and analysis*. Industrial Systems Research.

Abdelnour-Nocera, J. (2013). *Knowledge and technological development effects on organizational and social structures*. Information Science Reference. doi:10.4018/978-1-4666-2151-0

Acemoglu, D. (2012). Introduction to economic growth. *Journal of Economic Theory*, *147*(2), 545–550. doi:10.1016/j.jet.2012.01.023

Acemoglu, D., & Robinson, J. A. (2012). Why nations fail: The origins of power, prosperity, and poverty. London: Profile Books. doi:10.1355/ae29-2j

Altman, S. A. (2020, May 20). Will COVID-19 have a lasting impact on globalization? *Harvard Business Review*. https://hbr.org/2020/05/will-COVID-19-have-a-lasting-impact-on-globalization

Andrikopoulos, A., & Nastopoulos, C. (2015). *Κρίση και ρεαλισμός* [Crisis and Realism]. Propobos Publications.

Ansell, C., Sørensen, E., & Torfing, J. (2020). The COVID-19 pandemic as a game changer for public administration and leadership? The need for robust governance responses to turbulent problems. *Public Management Review*, *0*(0), 1–12. doi:10.1080/14719037.2020.1820272

Arbolino, R., & Caro, P. D. (2021). Can the EU funds promote regional resilience at time of COVID-19? Insights from the Great Recession. *Journal of Policy Modeling*, *43*(1), 109–126. doi:10.1016/j.jpolmod.2020.10.001

Azevêdo, D. G. (2020). Trade set to plunge as COVID-19 pandemic upends global economy. *WTO Trade Forecast Press Conference*. https://www.wto.org/english/news_e/pres20_e/pr855_e.htm

Bonilla-Molina, L. (2020). COVID-19 on route of the fourth industrial revolution. *Postdigital Science and Education*, *2*(3), 562–568. doi:10.100742438-020-00179-4

Boyer, R. (2019). The process of development: The contribution of régulation theory. In M. Nissanke & J. A. Ocampo (Eds.), *The Palgrave Handbook of Development Economics: Critical Reflections on Globalisation and Development* (pp. 177–227). Springer International Publishing. doi:10.1007/978-3-030-14000-7_6

Bragazzi, N. L. (2020). Digital technologies-enabled smart manufacturing and Industry 4.0 in the Post-COVID-19 Era: Lessons learnt from a pandemic. *International Journal of Environmental Research and Public Health*, *17*(13), 4785. doi:10.3390/ijerph17134785 PMID:32635191

Brinkman, R. (1995). Economic growth versus economic development: Toward a conceptual clarification. *Journal of Economic Issues*, *29*(4), 1171–1188. doi:10.1080/00213624.1995.11505746

Brown, L., Postel, S., & Flavin, C. (1992). From growth to sustainable development. In Population, technology, and lifestyle: The transition to sustainability (pp. 119–127). Island Press.

By, R. T. (2005). Organizational change management: A critical review. *Journal of Change Management*, *5*(4), 369–380. doi:10.1080/14697010500359250

Carmichael, J. (2011). *Leadership and management development*. Oxford University Press.

Chiras, D. D. (1995). Principles of sustainable development: A new paradigm for the twenty first century. *Journal of Environmental Science and Health. Part C, Environmental Carcinogenesis & Ecotoxicology Reviews*, *13*(2), 143–178. doi:10.1080/10590509509373475

Collin, P. H. (1998). *Dictionary of Medicine* (2nd ed.). Taylor & Francis.

Conceição, P. (2019). *Human development report 2019: Beyond income, beyond averages, beyond today: Inequalities in human development in the 21st century*. United Nations Development Programme.

Cox, M. (2017). The rise of populism and the crisis of globalization: Brexit, Trump and Beyond. *Irish Studies in International Affairs*, *28*, 9–17. doi:10.3318/isia.2017.28.12

Curran, L., & Eckhardt, J. (2020). Mobilizing against the antiglobalization backlash: An Integrated framework for corporate nonmarket strategy. *Business and Politics*, *22*(4), 612–638. doi:10.1017/bap.2020.9

Czifra, G., & Molnár, Z. (2020). COVID-19 and Industry 4.0. *Research Papers Faculty of Materials Science and Technology Slovak University of Technology*, *28*(46), 36–45. doi:10.2478/rput-2020-0005

de Geus, A. (1997). *The living company: Growth, learning and longevity in business*. Nicholas Brealey.

Deming, W. E. (1986). *Out of the crisis*. MIT Press.

Donaires, O. S., & Martinelli, D. P. (2019). Evolutionary management: From theory to action. *Systems Research and Behavioral Science*, *36*(1), 66–86. doi:10.1002res.2534

Endres, H. (2018). *Adaptability through dynamic capabilities: How management can recognize opportunities and threats*. Springer Gabler., doi:10.1007/978-3-658-20157-9

Engels, F. (1873). *Dialektik der natur (dialectics of nature)* (1972 edition). Progress Publishers.

European Council. (2020). *Special Meeting of the European Council (1 and 2 October 2020) – Conclusions*. Note from General Secretariat of the Council to Delegations. https://www.consilium.europa.eu/media/45910/021020-euco-final-conclusions.pdf

FAO. (2020). *COVID-19 and rural poverty*. Food and Agriculture Organization of the United Nations.

Fonseca, J. (2002). *Complexity and innovation in organizations*. Routledge., doi:10.4324/9780203279267

Fukuyama, F. (1992). *The end of history and the last man*. The Free Press.

Georgescu-Roegen, N. (1971). *The entropy law and the economic process*. Harvard University Press., doi:10.4159/harvard.9780674281653

Gillis, M., Perkins, D. H., Roemer, M., & Snodgrass, R. R. (1996). *Economics of development*. Norton.

Gorynia, M. (2020). Will COVID-19 Kill Globalization? In M. Marinov & S. T. Marinova (Eds.), COVID-19 and International Business: Change of Era (Chapter 6). Routledge. doi:10.4324/9781003108924-6

Gup, B. (Ed.). (2010). *The Financial and Economic Crises*. Edward Elgar Publishing., doi:10.4337/9781849806763

Hayes, J. (2018). *The theory and practice of change management*. Palgrave., doi:10.1057/978-1-352-00132-7

Hill, C., & Jones, G. R. (2004). *Strategic management theory: An integrated approach*. Houghton Mifflin.

Hodgson, G. (2002). Darwinism in economics: From analogy to ontology. *Journal of Evolutionary Economics*, *12*(3), 259–281. doi:10.100700191-002-0118-8

Hosseini, H. (2003). Why development is more complex than growth: Clarifying some confusions. *Review of Social Economy*, *61*(1), 91–110. doi:10.1080/0034676032000050329

Iacobucci, D., & Perugini, F. (2021). Entrepreneurial ecosystems and economic resilience at local level. *Entrepreneurship and Regional Development*, 1–28. doi:10.1080/08985626.2021.1888318

ILO. (2021a). *Global Wage Report 2020-21: Wages and minimum wages in the time of COVID-19*. International Labour Office.

ILO. (2021b). *Working from home: From invisibility to decent work*. International Labour Office.

IMF. (2020). *A crisis like no other, an uncertain recovery* (World Economic Outlook Reports, World Economic Outlook Update, June 2020). IMF.

International Civil Aviation Organization. (2020, November 12). *Effects of novel coronavirus (COVID-19) on civil aviation: Economic impact analysis.* Air Transport Bureau.

International Energy Agency. (2020). *World energy outlook 2020*. International Energy Agency.

Jaakkola, E. (2020). Designing conceptual articles: Four approaches. *AMS Review*, *10*(1), 18–26. doi:10.100713162-020-00161-0

Kelly, K. (1994). *Out of control: The new biology of machines, social systems and the economic world*. Basic Books.

Kotler, P., & Caslione, J. A. (2009). *Chaotics: The business of managing and marketing in the age of turbulence*. American Management Association.

Kotter, J. P. (1996). *Leading change*. Harvard Business School Press.

Krugman, P. R. (1996). *Development, geography, and economic theory*. MIT Press., doi:10.7551/mitpress/2389.001.0001

Lewin, K. (1943). Defining the "field at a given time.". *Psychological Review*, *50*(3), 292–310. doi:10.1037/h0062738

Liguori, E. W., & Pittz, T. G. (2020). Strategies for small business: Surviving and thriving in the era of COVID-19. *Journal of the International Council for Small Business*, *1*(2), 106–110. doi:10.1080/26437015.2020.1779538

Márquez-Ramos, L. (2021). Does digitalization in higher education help to bridge the gap between academia and industry? An application to COVID-19. *Industry and Higher Education*. Advance online publication. doi:10.1177/0950422221989190

Marx, K. (1867). *Das Kapital: Vol* (Vol. 1). Verlag von Otto Meisner.

Mason, C., & Hruskova, M. (2021). The impact of COVID-19 on entrepreneurial ecosystems. In Productivity and the Pandemic (pp. 59–72). Edward Elgar Publishing. doi:10.4337/9781800374607.00011

Matthewman, S., & Huppatz, K. (2020). A sociology of COVID-19. *Journal of Sociology (Melbourne, Vic.)*, *56*(4), 675–683. Advance online publication. doi:10.1177/1440783320939416

McKay, E. (2020). Luddism, isolation and education. *Journal for Activist Science and Technology Education*, *11*(2), 55–59. doi:10.33137/jaste.v11i2.34535

Meyer, C., & Davis, S. M. (2003). *It's alive: The coming convergence of information, biology, and business*. Crown Business.

Mhlanga, D., & Moloi, T. (2020). COVID-19 and the digital transformation of education: What are we learning on 4IR in South Africa? *Education Sciences*, *10*(7), 180. doi:10.3390/educsci10070180

Mill, J. S. (1848). *Principles of political economy*. Academic Press.

Mol, A., & Hardon, A. (2020). What COVID-19 may teach us about interdisciplinarity. *BMJ Global Health*, *5*(12), e004375. Advance online publication. doi:10.1136/bmjgh-2020-004375 PMID:33328203

Nelson, R., & Winter, S. (1982). *An evolutionary theory of economic change*. The Belknap Press of Harvard University Press.

Nnadozie, E., & Jerome, A. (2019). Definition and measurement of growth and development. In E. Nnadozie & A. Jerome (Eds.), *African Economic Development* (pp. 39–56). Emerald Publishing Limited., doi:10.1108/978-1-78743-783-820192003

OECD. (2018). *Development co-operation report 2018: Joining forces to leave no one behind*. OECD.

OECD. (2020). *E-commerce in the time of COVID-19* (Tackling Coronavirus (COVID-19): Contributing to a Global Effort). Secretary-General of the OECD.

Papanek, G. (2002). Economic growth versus economic development. *Periodica Polytechnica Social and Management Sciences*, *10*(2), 201–213.

Peet, R., & Hartwick, E. (2015). *Theories of development: Contentions, arguments, alternatives* (3rd ed.). Guilford Publications.

Peneder, M. (2017). Competitiveness and industrial policy: From rationalities of failure towards the ability to evolve. *Cambridge Journal of Economics*, *41*(3), 829–858. doi:10.2139srn.2714941

Perez, C. (1983). Structural change and assimilation of new technologies in the economic and social systems. *Futures*, *15*(5), 357–375. doi:10.1016/0016-3287(83)90050-2

Perroux, F. (1969). *L'économie du XXe siècle*. Presses Universitaires de France.

Puig, F., & Urzelai, B. (Eds.). (2019). *Economic clusters and globalization: Diversity and resilience*. Routledge., doi:10.4324/9780429444975

Raina, R. (2018). *Change management and organizational development*. SAGE Publications India.

Reiss, M. J. (2020). Science Education in the Light of COVID-19: The contribution of history, philosophy and sociology of science. *Science and Education*, *29*(4), 1079–1092. doi:10.100711191-020-00143-5 PMID:32836877

Rodríguez-Cohard, J. C., Juste-Carrión, J. J., & Vásquez-Barquero, A. (2020). Local development policies: Challenges for post-COVID-19 recovering in Spain. *Symphonya. Emerging Issues in Management*, *0*(2), 41–54. doi:10.4468/2020.2.05rodriguez.juste.vazquez

Ruttan, V. W. (1998). The new growth theory and development economics: A survey. *The Journal of Development Studies*, *35*(2), 1–26. doi:10.1080/00220389808422562

Sadeghloo, T., Sojasi, Q. H., Salehi, M., & Faal, J. A. (2018). Obstacles and methods of financing for the development of local entrepreneurship in Iran. *International Journal of Development Issues*, *17*(1), 114–138. doi:10.1108/IJDI-05-2017-0046

Sahasranamam, S. (2021). Innovation and entrepreneurship amdist coronavirus: A hybrid innovation network response. *South Asian Journal of Business Studies*. doi:10.1108/SAJBS-05-2020-0135

Savić, D. (2020). COVID-19 and Work from Home: Digital Transformation of the Workforce. *Grey Journal*, *16*(2), 101–104.

Scharmer, C. O., & von Ameln, F. (2019). Upgrading change management. *Gruppe. Interaktion. Organisation. Zeitschrift Für Angewandte Organisationspsychologie*, *50*(2), 193–196. doi:10.100711612-019-00464-2

Schumpeter, J. (1942). *Capitalism, socialism and democracy*. Harper & Brothers.

Schwab, K. (2016). *The fourth industrial revolution*. Crown Business.

Schwab, K., & Malleret, T. (2020). *COVID-19: The great reset*. World Economic Forum.

Senge, P. M. (1990). *The fifth discipline: The art and practice of the learning organization*. Doubleday/Currency.

Sharifi, S., & Claxton, J. (2015). Challenging notions of "Change" and "Change Management.". In F. Dievernich, K. Tokarski, & J. Gong (Eds.), *Change Management and the Human Factor: Advances, Challenges and Contradictions in Organizational Development* (pp. 209–222). Springer. doi:10.1007/978-3-319-07434-4_14

Smith, A. (1776). *An inquiry into the nature and causes of the wealth of nations*. doi:10.1093/oseo/instance.00043218

Soto-Acosta, P. (2020). COVID-19 pandemic: Shifting digital transformation to a high-speed gear. *Information Systems Management*, *37*(4), 260–266. doi:10.1080/10580530.2020.1814461

Szirmai, A., Naudé, W., & Goedhuys, M. (2011). *Entrepreneurship, innovation, and economic development*. Oxford University Press. doi:10.1093/acprof:oso/9780199596515.001.0001

Tobin, D. R., & Pettingell, M. S. (2008). *The AMA guide to management development*. AMACOM.

UNCTAD. (2020). Impact of the COVID-19 pandemic on global FDI and GVCs. UNCTAD.

UNESCO, UNICEF, & World Bank. (2020). What have we learnt? Overview of findings from a survey of ministries of education on national responses to COVID-19. UNESCO, UNICEF, World Bank.

United Nations. (2020a). Global Humanitarian Response Plan: COVID-19. *United Nations Coordinated Appeal*, (April – December), 2020.

United Nations. (2020b). The impact of COVID-19 on sport, physical activity and well-being and its effects on social development (Policy Brief No. 73). United Nations: Department of Economic and Social Affairs.

Verma, S., & Gustafsson, A. (2020). Investigating the emerging COVID-19 research trends in the field of business and management: A bibliometric analysis approach. *Journal of Business Research*, *118*, 253–261. doi:10.1016/j.jbusres.2020.06.057 PMID:32834211

Vlados, Ch. (2006). *Η δυναμική της παγκοσμιοποίησης και οι επιχειρήσεις στην Ελλάδα* [Dynamics of globalization and the enterprises in Greece]. Kritiki Publications.

Vlados, Ch. (2016). *Στρατηγική μικρομεσαίων επιχειρήσεων σε συνθήκες κρίσης: Η προσέγγιση Stra.Tech.Man* [Strategy of small and medium-sized enterprises: The Stra.Tech.Man approach]. Kritiki Publications.

Vlados, Ch. (2019a). Change management and innovation in the "living organization": The Stra.Tech.Man approach. *Management Dynamics in the Knowledge Economy*, *7*(2), 229–256. doi:10.25019/MDKE/7.2.06

Vlados, Ch. (2019b). The phases of the postwar evolution of capitalism: The transition from the current crisis into a new worldwide developmental trajectory. *Perspectives on Global Development and Technology*, *18*(4), 457–488. doi:10.1163/15691497-12341528

Vlados, Ch. (2020a). The growth and development analytical controversies in economic science: A reassessment for the post-COVID-19 era. *Journal of Economics and Political Economy*, *7*(4), 209–233.

Vlados, Ch. (2020b). The dynamics of the current global restructuring and contemporary framework of the US–China trade war. *Global Journal of Emerging Market Economies*, *12*(1), 4–23. doi:10.1177/0974910119896636

Vlados, Ch. (2021). The Stra.Tech.Man Scorecard. *International Journal of Business Administration*, *12*(2), 36–63. doi:10.5430/ijba.v12n2p36

Vlados, Ch., & Chatzinikolaou, D. (2019a). Business ecosystems policy in Stra.Tech.Man terms: The case of the Eastern Macedonia and Thrace region. *Journal of Entrepreneurship. Management and Innovation*, *15*(3), 163–197. doi:10.7341/20191536

Vlados, Ch., & Chatzinikolaou, D. (2019b). Crisis, institutional innovation and change management: Thoughts from the Greek case. *Journal of Economics and Political Economy*, *6*(1), 58–77.

Vlados, Ch., & Chatzinikolaou, D. (2020). Thoughts on competitiveness and integrated industrial policy: A field of mutual convergences. *Review of World Economics*, *11*(3), 12–25. doi:10.5430/rwe.v11n3p12

Vlados, Ch., Chatzinikolaou, D., Katimertzopoulos, F., & Koutroukis, T. (2019). Regional underdevelopment and less developed business ecosystems: The case of Eastern Macedonia and Thrace. *Bulletin of Applied Economics*, *6*(2), 31–44.

Vlados, Ch., Deniozos, N., & Chatzinikolaou, D. (2018). Towards a new approach of local development under crisis conditions: Empowering the local business ecosystems in Greece, by adopting a new local development policy. *International Journal of Regional Development*, *5*(1), 1–24. doi:10.5296/ijrd.v5i1.11955

Wang, G. G., Korte, R. F., & Sun, J. Y. (2008). Development economics: A foundation for HRD policy studies in developing countries. *Advances in Developing Human Resources*, *10*(6), 848–862. doi:10.1177/1523422308324666

Wen, J., Wang, W., Kozak, M., Liu, X., & Hou, H. (2020). Many brains are better than one: The importance of interdisciplinary studies on COVID-19 in and beyond tourism. *Tourism Recreation Research*, *1–4*. Advance online publication. doi:10.1080/02508281.2020.1761120

Winarsih, I. M., & Fuad, K. (2021). Impact of COVID-19 on Digital Transformation and Sustainability in Small and Medium Enterprises (SMEs): A Conceptual Framework. In L. Barolli, A. Poniszewska-Maranda, & T. Enokido (Eds.), Complex, Intelligent and Software Intensive Systems (pp. 471–476). Springer International Publishing. doi:10.1007/978-3-030-50454-0_48

World Bank. (2020a). *World Bank Predicts Sharpest Decline of Remittances in Recent History*. World Bank, Press Release.

World Bank. (2020b). *Global Economic Prospects*. World Bank. doi:10.1596/978-1-4648-1553-9

World Economic Forum. (2020). *The Future of Jobs Report*. World Economic Forum.

Xu, L., & Liu, J. C. (2017). Growth vs. development: A challenge to China for the 21st century. In *Economic Globalization in Asia* (pp. 146–155). Routledge.

Yaeger, T. F., & Sorensen, P. F. (2009). *Strategic organization development: Managing change for success*. Information Age Pub.

Zeleny, M. (1980). *Autopoiesis: A theory of living organizations*. Elsevier Science Ltd.

Zuvekas, C. (1980). *Economic Development*. Macmillan International Higher Education.

ADDITIONAL READING

Bennett, E. E., & McWhorter, R. R. (2021). Virtual HRD's role in crisis and the Post COVID-19 professional lifeworld: Accelerating skills for digital transformation. *Advances in Developing Human Resources*, *23*(1), 5–25. doi:10.1177/1523422320973288

Bianchi, P., & Labory, S. (2019). *Industrial policy for the manufacturing revolution: Perspectives on digital globalization*. Edward Elgar Publishing. doi:10.4337/9781786430328

Chatzinikolaou, D., & Vlados, Ch. (2019). Evolutionary economics and the Stra.Tech.Man approach of the firm into globalization dynamics. *Business. Management and Economics Research*, *5*(10), 146–160. doi:10.32861/bmer.510.146.160

Doucet, A., Evers, J., Guerra, E., Lopez, N., Soskil, M., & Timmers, K. (2018). *Teaching in the Fourth Industrial Revolution Standing at the Precipice*. Routledge. doi:10.4324/9781351035866

Kelly, R. (2019). *Constructing LEADERSHIP 4.0: Swarm Leadership and the Fourth Industrial Revolution*. Springer International Publishing. doi:10.1007/978-3-319-98062-1

Nelson, R., Dosi, G., Helfat, C., Winter, S., Pyka, A., Saviotti, P., Lee, K., Malerba, F., & Dopfer, K. (2018). *Modern evolutionary economics: An overview*. Cambridge University Press. doi:10.1017/9781108661928

Roth, S. (2021). The Great Reset. Restratification for lives, livelihoods, and the planet. *Technological Forecasting and Social Change*, *166*, 120636. Advance online publication. doi:10.1016/j.techfore.2021.120636

Vlados, Ch., Deniozos, N., & Chatzinikolaou, D. (2019). *Global crisis and restructuring: Theory, analysis, and the case of Greece*. KSP Books. http://books.ksplibrary.org/978-605-7602-89-3/

KEY TERMS AND DEFINITIONS

Adaptability: The adaptability concept refers to socioeconomic organizations' ability to transform their internal environment by building upon their comparatively competitive advantages to insert themselves into their co-evolving external environment.

Change Management: Change management is the specific management approach that investigates how socioeconomic organizations manage to overcome resistance to change that irreversibly appears in their internal environment and adapt to the external environment's transformations.

Economic Growth: Economic growth means the increase in one or more quantitative dimensions of a socioeconomic organization.

Evolutionary Management: Evolutionary management is the specific approach to managing resources where the various socioeconomic systems are examined as complex entities that adapt themselves to their external environment evolutionarily via adaptation and innovation actions.

Fourth Industrial Revolution: The fourth industrial revolution refers to the historical phase of global capitalism where radical transformations occur at institutional and socioeconomic levels due to the diffusion of "cyber-physical" systems.

Innovation: Innovation is the dynamic phenomenon that produces new and more effective solutions than in the past in both newer and older problems, leading to performance improvement in all or parts of organizational aspects.

Interdisciplinarity: Interdisciplinarity involves understanding all socioeconomic phenomena by examining contributions from all scientific fields that concern the specific case under investigation.

Post-COVID-19 Era: Post-COVID-19 era signals the epoch that arises after the global health and socioeconomic crisis of the COVID-19 pandemic that accelerates the developments towards the fourth industrial revolution.

Socioeconomic Development: Socioeconomic development is the progressive reinforcement of a socioeconomic organization's quantitative and qualitative dimensions towards a higher level of efficiency, well-being, justice, and democracy at all levels.

Stra.Tech.Man Approach (Strategy-Technology-Management Synthesis): The Stra.Tech.Man approach is about how all socioeconomic organizations synthesize their internal physiological spheres of strategy, technology, and management to innovate, survive, and develop.

ENDNOTE

[1] Of course, there are a plethora of dynamic models nowadays that are called upon to offer more comprehensive responses and, without a doubt, if an economic growth model is wrong, this can be improved with a better mathematical model (this is the way all models are generated), without implying that mathematical thought is inadequate altogether. If a model is static, someone might try to turn it into a dynamic one. If a model is linear, someone might try making it non-linear. If a model has only one variable, someone might add more ("denying" mathematics is not self-evident).